

Adaptive Design of Clustered Surveys and Experiments



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Introduction and Motivation

State Problem

Algorithms and Performance Guarantees

Inference

Application

Conclusion

- ▶ Design of clustered experiments: clusters (K , **width**) and units/cluster (N , **depth**)
 - ▶ select (K, N) to **minimize** the **variance** of a population mean or ATE
 - ▶ subject to a **hard budget**, **variable cost** per unit, and **fixed cost** per cluster—sunk before recruitment like travel or staff training

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- ▶ Optimal design is known since Cochran (1977) which depends on the **ICC**
 - ▶ Relative correlation within clusters, but **unknown**
- ▶ In practice, designers **guess** ...
 - ▶ ...**guessing is expensive**, the excess variance of ICC misimputation is of the same order as the target variance
 - ▶ ...**guessing is difficult**, ICC varies an order of magnitude within a single population (Knox and Chondros, 2004)
- ▶ Adaptive designs are enticing but they **mess up inference**
 - ▶ why minimize estimator's variance if we cannot use standard inference?

- ▶ **Research questions**

- ▶ Can we design optimal clustered experiments under unknown ICC?
- ▶ Can we do so without compromising inference?

- ▶ Two **pilot-based algorithms** learn the ICC while the experiment runs

- ▶ induce negligible excess variance, beat any static design, and rate unimprovable

- ▶ Under **error symmetry, adaptivity does not compromise inference**

- ▶ **general theory** of inference invariance which includes Neyman allocations

- ▶ **End to end illustration**

- ▶ simulations, fielded a **survey** in real time using the algorithm, revisit Fink et al. (2020, AER)—up to **41% lower variance** at equal cost

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- ▶ **In practice**, plug-in ICC guesses
- ▶ **Survey Theory**
 - ▶ algorithmically-driven literature (Bethel, 1989; Barcaroli et al., 2022)
 - ▶ central role of ICC, but they rarely estimate it **inside** the optimization problem
- ▶ **Statistics and Biostatistics**
 - ▶ **Static minmax** approach over ICC uncertainty (van Breukelen and Candel, 2015)
 - ▶ adaptive designs can dominate every static policy by an order of magnitude
 - ▶ **Sample-size re-estimation** with nuisance parameters (Stein, 1945)
 - ▶ ICC re-estimation (Lake et al., 2002; van Schie and Moerbeek, 2014) no cost structure, no optimality theory, a single margin (width or depth)
- ▶ First paper to establish optimality guarantees against the oracle under unknown ICC in the joint allocation problem

- ▶ Extends mechanically to experimental design
- ▶ Nested **random effects** model [A1]

$$y_{ki} = \mu + u_k + \varepsilon_{ki}, \quad \text{Var}(u_k) = \tau^2, \quad \text{Var}(\varepsilon_{ki}) = \sigma^2, \quad u_k \perp\!\!\!\perp \varepsilon_{ki}$$

- ▶ **unknown variance ratio** $\lambda_0 = \sigma^2/\tau^2$, one-to-one with ICC = $1/(1 + \lambda_0)$
- ▶ $\omega = (\tau^2, \lambda_0) \in \text{int}(\Omega)$ with Ω **known** and **compact** [A2]
- ▶ **Goal**

$$\min_{K,N} \text{Var}(\hat{\mu}_{K,N}) = W(K, N) \quad \text{st} \quad K(F + VN) \leq B \quad \text{a.s.}$$

- ▶ F fixed cost per cluster, V variable cost per unit, B fixed budget

$$N^*(\lambda_0) = \sqrt{\lambda_0 F/V}, \quad K^*(\lambda_0) = \frac{B}{F + V N^*(\lambda_0)}$$

- Assume **interior solution** for $\lambda \in [\lambda_m, \lambda_M]$ [**A3**]

$$W_0 = W(K^*(\lambda_0), N^*(\lambda_0)) = \Theta(1/B)$$

- **Excess variance** (Neyman Regret)

$$\Delta_\pi = W(K^\pi, N^\pi) - W_0$$

- **Cost of guessing** (Lemma 5.2)

$$\Delta_{\text{guessing}} = W(K^*(\lambda), N^*(\lambda)) - W_0 = \Theta(1/B) \text{ for all } \lambda \neq \lambda_0$$

- **same order as the oracle variance itself**
- true for **any** static input, including the minimax λ input (van Breukelen and Candel, 2015)

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► **Pilot-based policies**

$$\text{pilot } (K_p, N_p) \implies \hat{\lambda}_p = \hat{\lambda}_{K_p, N_p} \implies (K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$$

- clusters and units cannot be *unsampled*
- **implementable** pilots: $K_p \leq K^*(\hat{\lambda}_p)$ and $N_p \leq N^*(\hat{\lambda}_p)$ a.s. $\implies$ **circularity**

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- ▶ **Why committing early?**

- ▶ suppose $K_p = 20$ and $K^*(\hat{\lambda}_p) = 50 \implies$ additional clusters are valuable for λ estimation

► Pilot-based policies

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► EtC-CS: two stages with **smallest pilot**

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► Why committing early?

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► **ACS**: fixes shallow $N_p = N^*(\lambda_m)$, samples **one cluster at a time**, stops before **oversampling** $s = \inf\{k : k + 1 > K^*(\hat{\lambda}_k)\}$

► nearly **all exploration inside the pilot**, at no additional sampling cost

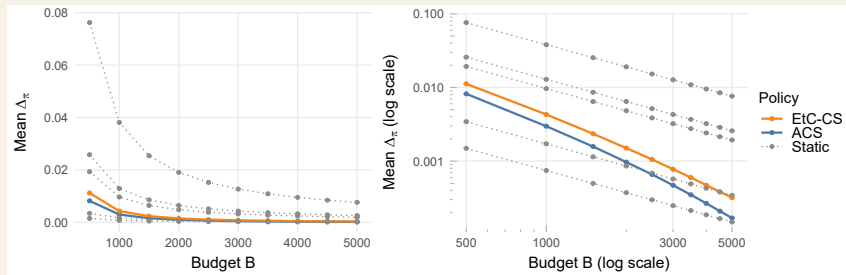
Theorems 5.4 and 5.5. Upper and Lower Bounds

Under $[A1 - A3] + \text{subgaussian } u_k, \varepsilon_{ki}$ $[A4]$, there exist constants $C_\pi < \infty$, $\perp B$,

$$\mathbb{E} [\Delta_{\text{EtC-CS}}] \leq \frac{C_{\text{EtC-CS}}}{B^2}, \quad \mathbb{E} [\Delta_{\text{ACS}}] \leq \frac{C_{\text{ACS}}}{B^2}$$

Moreover, for every policy π there exists $\omega \in \Omega$ such that $\mathbb{E}_\pi[\Delta_\pi] \geq C/B^2$.

- ▶ **asymptotically negligible** excess variance, $O(1/B^2) \ll \Theta(1/B)$
- ▶ algorithms asymptotically **dominate every misspecified static policy**
- ▶ **minimax rate optimal**, matching bounds



Notes: Static policies at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $F = 20$, $V = 2$.

- **Low excess variance** with ACS outperforming EtC-CS
- **Faster convergence**, slopes ≈ -2 (EtC-CS and ACS), ≈ -1 (static)

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- ▶ (K^π, N^π) **depends on pilot data** $\Rightarrow$ is $\hat{\mu} \mid (K^\pi, N^\pi)$ centered or asympt normal?
 - ▶ adaptive sampling biases estimates and destroys normality (Nie et al., 2018; Wager, 2024)
- ▶ **Adjustment tradition:** restore normality by **reweighting** the observations
 - ▶ Hadad et al. (2021); Zhang et al. (2021); Yin and Shen (2005)
- ▶ **Invariance tradition:** under **exact normality** designs driven by **sample variances** do not disturb inference on means
 - ▶ Stein (1945); Rinott (1978); Dudewicz and Dalal (1975)

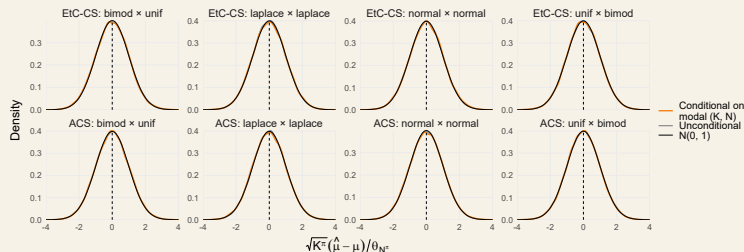
- ▶ This paper builds on the **invariance tradition** ...
- ▶ **reduces** the policy class to policies reading **squared errors** – **even**
- ▶ **extends** the invariance principle in three ways
 - ▶ **normality** → **symmetry** of the errors
 - ▶ **sample variance** → any **asymptotically even** functional of the data
 - ▶ **fixed-stage** adaptivity → **arbitrary** sequential designs
- ▶ Theory **extends well beyond** my algorithms and the cluster setting
 - ▶ **Neyman allocations** and **Optimal propensity scores** (Hahn et al., 2011)
 - ▶ allows conditioning on **non-degenerate events** (Hahn et al., 2011) and avoids **pilot data exclusion** (Viviano, 2022)

Theorem 6.10. Design-Robust Inference for EtC-CS and ACS

Under $[A1 - A3] + (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots) \stackrel{d}{=} -(u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ [A5], for any design event A_B satisfying an anti-concentration condition,

$$\mathcal{L} \left(\frac{\sqrt{K^\pi} (\hat{\mu}_{K^\pi, N^\pi} - \mu)}{\hat{b}_{K^\pi, N^\pi}} \mid A_B \right) \implies N(0, 1) \quad \pi \in \{\text{EtC-CS}, \text{ACS}\}, \quad \mathbb{P}(A_B) > 0$$

- **Nothing needs to change:** full sample, usual clustered SEs and normal CV—no reweighting, no sample splitting, no discarded data



Notes: Empirical density at the modal design. Distribution of u_k (left), of ε_{ki} (right). Distributions considered: Bimodal-Uniform, Laplace-Laplace, Normal-Normal, Uniform-Bimodal. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 2,000$, $F = 20$, $V = 2$.

- Conditional and unconditional laws **coincide**
- **Boundary is real**, odd policies or skewed errors introduce **not vanishing** distortions

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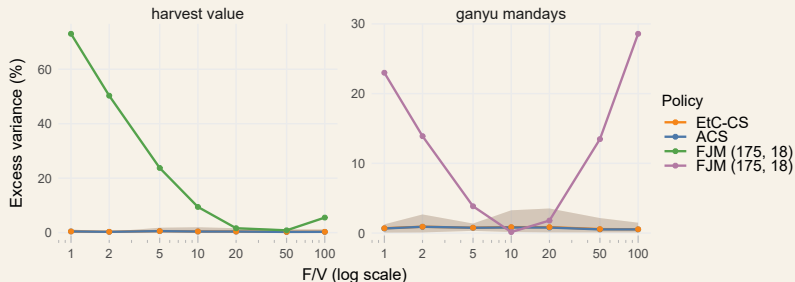
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- ▶ **Setting:** loans to farmers in Zambia with treatment at the **village** level
 - ▶ $K = 175$ villages $\times N \approx 18$ households
- ▶ **Was the realized design cost-optimal?**
 - ▶ do not observe cost structure, but possibly $F/V \in [1, 2]$
 - ▶ their design is only rationalizable at $F/V \approx 33$
 - ▶ away from this ratio, the realized design carries **up to 70% excess variance**



Notes: Excess variance is presented relative to the oracle design (%). Lines are means over 10,000 simulations, bands are 95% intervals across simulations.

- ▶ Algorithms remove nearly all excess variance uniformly across cost-ratios
- ▶ Not claiming that FJM got it wrong—simply **no design is safe without cost-aware learning**
- ▶ **Symmetry is testable**, and not rejected here, so standard inference applies

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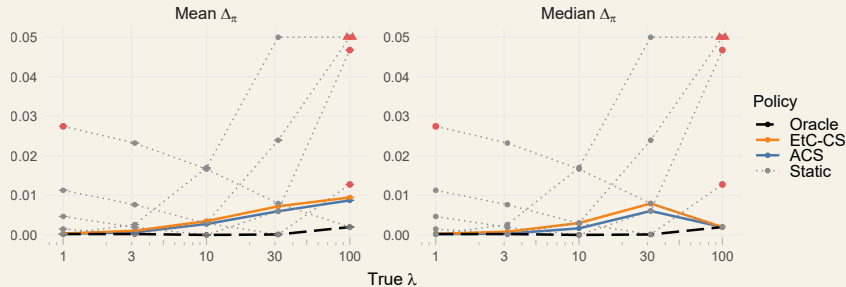
Conclusion

- ▶ **Framework** for cost-optimal clustered designs under **unknown ICC**, casting the classical width-vs-depth trade-off into a **sequential learning** problem
- ▶ Two **adaptive algorithms**
 - ▶ excess variance $O(1/B^2)$, negligible against the $\Theta(1/B)$ oracle variance
 - ▶ rate unimprovable, and dominate any misspecified static design $\Theta(1/B)$
- ▶ **Adaptivity does not compromise inference under symmetry**
 - ▶ even designs leave estimators asymptotically normal
 - ▶ no corrections, no discarded data
 - ▶ general principle which validates many popular sampling rules
- ▶ **Economically meaningful**
 - ▶ runs live in the field
 - ▶ 41% lower variance in FJM at equal cost,

Thank you!

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Carlos Gonzalez-Perez — and I am on the market!



Notes: Static policies at $\lambda \in \{1, 3.16, 10, 31.6, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $B = 2,000$, $F = 20$, $V = 2$.

- Adaptive **worst case beats every static worst case** (minimax included), static policies win only near their guess and only at finite budgets

- ▶ Under symmetry, cluster errors **factor** into

signs $\times$ magnitudes with signs $\perp\!\!\!\perp$ magnitudes

- ▶ Even functionals of the data are functions of **magnitudes alone**
 - ▶ conditioning on an even design **freezes** the design and the magnitudes...
 - ▶ ...but leaves the signs **untouched**, so a **CLT runs through the signs**
- ▶ **Sharp boundary**
 - ▶ reading **odd** functionals (interim means, futility rules) or **asymmetric** errors restores the usual pathologies

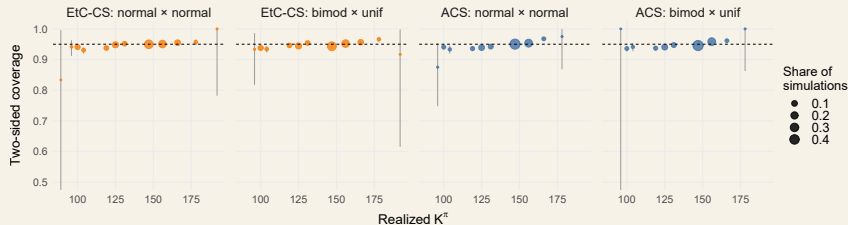
Policy Reads...	Assumption on the Errors	
	Gaussian	Symmetry (New)
Levels	✗	✗
Residuals	✓	✗
Squared Errors (even)	✓	✓
Squared Residuals (asympt even)	✓	✓

Table: Policies and Error Assumptions

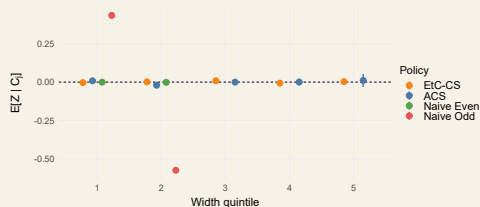
Scale	Does Surrogate Coupling Survive?	Non-degenerate Events?	Tools
Unit	✗	—	—
$1/B$	✓	✗	Anscombe
$1/\sqrt{B}$ (new)	✓	✓	Symmetry and Evenness

Table: Scale of Conditioning Events

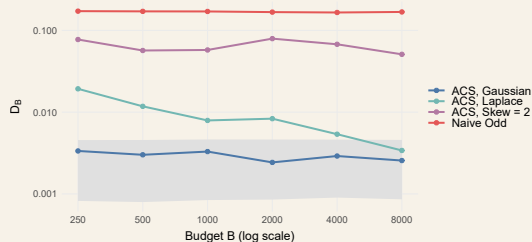
- ▶ We can condition wherever surrogate coupling survives
- ▶ The symmetry $\times$ evenness theory allows for conditioning on **non-degenerate** events
- ▶ Unconditional inference does **not** require symmetry (Anscombe)



Notes: Design-conditional coverage of $\hat{\mu}_{K_B^\pi, N_B^\pi} \pm 1.96 \hat{b}_{K_B^\pi, N_B^\pi} / \sqrt{K_B^\pi}$. Grey lines are Clopper–Pearson 95% intervals for each realized K_B^π . 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 5,000$, $F = 20$, $V = 2$.



(a) $E[Z_B^\pi | C_j]$



(b) $D_B = \sup_{t \in \mathcal{T}} \max_j \left| \mathbb{E}[e^{itZ_B^\pi} \mathbb{1}_{C_j}] - \phi(t)\mathbb{P}(C_j) \right|$

Notes: $(C_j)_j$ are quintiles of the realized K^π . Grey band represents worst-case 95% range of D_B over B under exact conditional normality. Symmetric errors are $N(0, 1)$, asymmetric errors are centered $\Gamma(1, 1)$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $B = 2,000$, $F = 20$, $V = 2$.

► **Asymmetry or Policy-Oddness compromise inference**